Jerónimo García-Mejía

Personal Data

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Website:	jeronimomaths.github.io
Research Interests:	Geometric Group Theory and Geometric Topology.

Academic Positions

Oct 2025 – Present	University of Warwick Research Fellow Mathematical Institute Postdoctoral Mentor: Dr. Ric Wade
Dec 2024 – Oct 2025	University of Oxford Postdoctoral Research Associate Mathematical Institute Postdoctoral Mentor: Dr. Ric Wade

Education

Sep 2024	Karlsruhe Institute of Technology PhD in Mathematics Thesis: *Dehn functions of groups with filiform subgroups* Advisor: Jun.-Prof. Claudio Llosa Isenrich
Aug 2020	University of Bonn M.Sc. in Mathematics Thesis: *Group actions on hyperbolic spaces* Advisor: Prof. Ursula Hamenstädt
Apr 2018	Universidad Nacional Autónoma de México B.Sc. in Mathematics Thesis: *Construction of classifying spaces* (in Spanish) Advisor: Prof. Carlos Prieto

Scholarships

2018–2020	Rosa Luxemburg Stiftung (Rosa Luxemburg Foundation) Scolarship

Publications

1. Yu-Chang Chan, Jerónimo García-Mejía, and Matteo Migliorini, *Complete classification of the Dehn functions of Bestvina–Brady groups.*
Geometric and Functional Analysis (2026).

2. Jerónimo García-Mejía, Claudio Llosa Isenrich and Gabriel Pallier *On Dehn functions of central products of nilpotent groups.*
Transactions of the American Mathematical Society (2026).

3. Jerónimo García-Mejía and Antoine Goldsborough, *Random Dehn function of groups.*
Journal of Topology and Analysis (2026).

Preprints

1. Jerónimo García-Mejía, and Matteo Migliorini, *Conjugator length of Bestvina–Brady groups* . (Nov. 2026), **arXiv:2611.05619**.

Other work containing original research

- *Dehn functions of groups with filiform subgroups.*
  PhD thesis, Karlsruhe Institute of Technology.
- *Group actions on hyperbolic spaces.*
  Master thesis, University of Bonn. Unpublished.

Invited conference talks

Oct 2026	*Distortion and coarse median preserving automorphisms of special groups*, Groups and Geometry in the South East Meeting, University College London. (scheduled).
Oct 2026	*Complete classification of the Dehn functions of Bestvina–Brady groups*, Conference: LiT IV Landing in Topology, Università di Pisa (scheduled).
Sep 2026	*Distortion and coarse median preserving automorphisms of special groups*, Conference: Automorphisms groups in Bonn, Universität Bonn.
Jun/Jul 2026	*La geometría del problema de la palabra* Summer School: Escuela de Teoría Geométrica de Grupos, Instituto de Matemáticas Unidad Oaxaca, México.

Invited seminar talks

May 2026	*Complete classification of the Dehn functions of Bestvina–Brady groups*, Geometry and Topology Seminar, University of Glasgow.
Apr 2026	*Complete classification of the Dehn functions of Bestvina–Brady groups*, Oberseminar Geometric Group Theory, Universität Bonn.
Jan 2026	*Complete classification of the Dehn functions of Bestvina–Brady groups*, Topology Seminar, University of Oxford.
Jan 2026	*Complete classification of the Dehn functions of Bestvina–Brady groups*, Geometry and Topology Seminar, University of Warwick.
Nov 2025	*Complete classification of the Dehn functions of Bestvina–Brady groups*, Algebra Seminar, University of Bristol.
May 2025	*Dehn functions of Bestvina–Brady groups*, Geometric Group Theory Seminar, University of Cambridge.
July 2024	*Dehn functions of nilpotent groups*, Geometric Group Theory Seminar, University of Oxford.
Apr 2024	*Dehn functions of nilpotent groups*, Research seminar Model theory and groups, Universität Münster.
Mar 2024	*Funciones de Dehn de grupos nilpotentes*, Long-distance seminar in Geometric Group Theory in Mexico.
Jan 2024	*Dehn functions of central products of nilpotent groups*, Geometry and Dynamics Seminar, Université de Lille.
Oct 2022	*The Geometry of the Word Problem*, Baby Geometri Seminar, Università di Pisa - Scuola Normale Superiore
May 2020	*Group actions on hyperbolic spaces*, Junior Geometry Seminar, Universität Heidelberg.

Contributed talks

May 2026	*Lightning Talk*, IMBM Istanbul Topology Days:: A Conference on Developments in Contact Topology, Mapping Class Groups, and Beyond, Boğaziçi University, Istanbul, Turkey.
May 2023	*Lightning Talk*, Conference Geometry of subgroups, Thematic program on Geometric Group Theory, Centre de Recherches Mathématiques, Montréal.

Feb 2023	*Lightning Talk*, Young Geometric Group Theory XI, Münster.
Feb 2021	*Group actions, hyperbolic spaces and small cancellation*, AG Seminar Metric Geometry, Karlsruhe Institute of Technology, Karlsruhe.

Conferences and workshops attendance

May 2026	*IMBM Istanbul Topology Days: A Conference on Developments in Contact Topology, Mapping Class Groups, and Beyond*, Boğaziçi University, Turkey.
Mar 2026	*Topological and Homological Methods in Group Theory 2026*, Bielefeld University, Germany.
Mar 2026	*Journées Trans-Couesnon de théorie des groupes*, University of Caen, France.
Sep 2025	*Actions on graphs and metric spaces* part of the program Operators, Graphs, Groups, Isaac Newton Institute for Mathematical Sciences, United Kingdom.
Jul 2025	*Non-positive curvature and applications*, Isaac Newton Institute for Mathematical Sciences, United Kingdom.
Jun 2025	*The William Rowan Hamilton Geometry and Topology Workshop: Celebrating Martin Bridson's 60th Birthday*, The Hamilton Mathematics Institute, Trinity College Dublin, Ireland.
May 2025	*Groups and Geometry in South England*, University of Oxford, United Kingdom.
Aug 2024	*Workshop on Profinite Rigidity*, Karlsruhe Institute of Technology, Germany.
Apr 2024	*Young Geometric Group Theory XII*, University of Bristol, United Kingdom.
Mar 2024	*Topological and Homological Methods in Group Theory 2024*, Bielefeld University, Germany.
Jun 2023	*Cube complexes and combinatorial group theory* part of the Thematic program on Geometric group theory, Centre de Recherches Mathématiques, Canada.
Jun 2023	*Groups around 3-manifolds* part of the Thematic program on Geometric group theory, Centre de Recherches Mathématiques, Canada.
May 2023	*Geometry of subgroups* part of the Thematic program on Geometric group theory, Centre de Recherches Mathématiques, Canada.
Feb 2023	*Young Geometric Group Theory XI*, Münster University, Germany.
Jul 2022	*Metric Geometry and Geometric Analysis*, MSRI-Oxford Summer Graduate School, Oxford University, United Kingdom.
Jun 2022	*Hyperbolic groups and their generalisations*, Institut Henri Poincaré, France.
Mar 2022	*Topological and Homological Methods in Group Theory 2022*, Bielefeld University, Germany.
Feb 2022	*Workshop on finiteness properties*, Karlsruhe Institute of Technology, Germany.

Extended visits

Oct 2025 – Jan 2026	University of Oxford, Mathematical Institute Academic visitor
Sep 2025	Isaac Newton Institute Semester program *Operators, Graphs, Groups*
May – Jun 2023	Centre de Recherches Mathématiques Thematic program on *Geometric group theory*

Teaching experience

My teaching experience as *Teacher's Assistant* covers preparing exercise-sheets, presenting solutions in front of the class, and grading. All of this in Spanish, English, and German.

Winter 2021/22	*Elementare Geometrie (Elementary Geometry)*, Bachelor course in German at the Karlsruhe Institute of Technology.
Summer 2021	*Geometric Group Theory*, Master course in English at the Karlsruhe Institute of Technology.

Winter 2020/21	*Proseminar Geometrie: Knotentheorie (Knot theory)*, Bachelor Seminar in German at the Karlsruhe Institute of Technology.
Winter 2017/18	*Analytical Geometry I* and *Topology I*, Bachelor courses in Spanish at Universidad Nacional Autónoma de México.

Supervision experience

Jul – Sep 2026	I supervised a 3rd year student as part of the *Undergraduate Research Support Scheme* at the University of Warwick. The project was titled *Cops and Robbers in Geometric Group Theory*. It consisted on determining the "strong cop number" for groups.
Jul – Aug 2026	I supervised a group of three students as part of the *Summer Research programme for female-identifying students* at the University of Warwick. The project was title *The geometry and combinatorics of geodesics in nilpotent groups*. It consisted on constructing explicit combings for nilpotent groups and abelian-by-abelian groups.

Services

Present	Organiser of the Warwick Geometry and Topology Seminar.
Aug 2024	I co-organised of the *workshop Profinite Rigidity*, joint with the University of Münster, Regensburg University, and the Karlsruhe Institute of Technology, Germany.
Apr 2022	I co-organised of the *Seminar Group (co)homology* of the "RTG Asymptotic Invariants and Limits of Groups and Spaces" joint with the Karlsruhe Institute of Technology and Heidelberg University, Germany.
Oct 2021 – Sep 2022	I was a speaker of the RTG "Asymptotic Invariants and Limits of Groups and Spaces", Germany.